Teacher Wu Kuan | See It at a Glance #13

Parents must see **7th grade · 1st midterm** English lecture notes matching the Chinese handout for -3^2 vs $(-3)^2$

1. The trap question

$$-3^2 = ?$$

Right before seventh-grade midterms, kids lose points not on the hard problems — but on this one.

2. The most common mistake

$$-3^2 = ?$$

9 **−9**

Many kids write 9.

No parentheses -3^2

Negative sign is outside. The square only applies to 3.

First compute $3^2 = 9$, then put the negative sign in front → **−9**.

$-3^2 = -(3^2) = -9$

Adding a negative sign = flip to the other side of 0 on the number line.

With parentheses $(-3)^2$

Parentheses enclose the negative sign. The square applies to the whole (-3).

$(-3) \times (-3) = 9$

$(-3)^2 = 9$ · two negatives make a positive

3. Parent check method

Have your child write both side by side and say **what the square applies to**. If they can explain it, they've got this down.

-3^2

Square acts on **3**

= −9

$(-3)^2$

Square acts on **(−3)**

= 9

Key takeaway (from the video)

- -3^2 has **no parentheses**: square only 3 → then add the minus → **−9**.
- $(-3)^2$ has **parentheses**: minus is inside → $(-3)\times(-3)$ → **9**.
- Check: "Who does the square act on?"

Practice (same idea as the Chinese notes / video)

1. Compute -5^2.
2. Compute $(-5)^2$.
3. Compute -2^2 and $(-2)^2$. Are they equal?
4. True or false: $-4^2 = 16$. Explain.
5. Fill in: $-7^2 = -(____) = ____$; $(-7)^2 = (____)\times(____) = ____$.

Saber Math · Teacher Wu Kuan (吳寬) · See It at a Glance #13
Companion to YouTube Short RHF4f_NM5eI · English handout · Do not publish until approved
Weekly practice trial: https://kuan-weekly.vercel.app/sample

Answer key

See It at a Glance #13 — English lecture notes

1. $-5^2 = -(5^2) =$ **-25**
2. $(-5)^2 = (-5)\times(-5) =$ **25**
3. $-2^2 =$ **-4**; $(-2)^2 =$ **4**. **Not equal.**
4. **False.** $-4^2 = -(4^2) = -16$, not 16. (Only $(-4)^2 = 16$.)
5. $-7^2 = -($**7^2**$) =$ **-49**; $(-7)^2 = ($**-7**$)\times($**-7**$) =$ **49**.
